Reading the Economy from Space

Text- and Night-Light-Guided Satellite Embeddings for Municipal GDP in Minas Gerais, Brazil

Group 2: Hemant (22CS30029), Saras Dipak Wagh (22CS30048), Pallav Agarwal (22CS30040)

Term project, Geospatial Reasoning using LLMs and Generative AI · github.com/chadsaras/GIS_Project

October 2026

Abstract

We estimate 2022 GDP per capita for all 853 municipalities of Minas Gerais, Brazil, from free satellite data on a 1 km grid (575,166 cells). The pipeline combines two recent ideas that had not been tested together: aligning Google's frozen satellite embeddings (GSED) with text written by a vision–language model (as in UrbanCLIP), and steering the embeddings towards economic activity with VIIRS night lights (as in Yue, Zhao and Hu, 2026). Cell outputs are pooled to municipalities and trained on official IBGE GDP. Every number is out-of-sample under 5-fold spatial cross-validation over IBGE's 70 immediate regions, with all three learning stages retrained in each round. The best model, a validation-chosen blend of the best embedding model with the night-light benchmark, reaches R^2 0.52 (RMSE 0.41 in log GDP per capita), against 0.27 (0.51) for night lights alone and 0.39 (0.47) for LightGBM on the same embeddings. Adding captions gives the largest single gain (RMSE -0.025, 95% block-bootstrap interval $[-0.050, +0.004]$), but none of the four planned comparisons (night-light steering, text, POIs and roads, hybrid) is significant. The largest errors are mining and steel municipalities, where a few cells produce most of the output, and small public-sector towns. A blind audit of 300 captions finds a false claim in 35% of raw captions and 23% after our data-based cleaning.

1 Introduction

Official GDP below the national level is often late, estimated from partial data, or missing. Satellites observe every place with the same instruments every year, so a model that reads economic activity from satellite data gives an independent and cheap estimate for every district. Two lines of recent work point in different directions. *UrbanCLIP* (Yan et al., 2024) generates a caption for every satellite tile with a vision–language model and aligns image and text, reporting gains on urban indicators. *Estimating Economic Activity from Satellite Embeddings* (Yue, Zhao and Hu, 2026) starts from Google's 64-dimensional annual satellite embeddings and steers them with night lights before aggregating to regions.

This project combines both in one pipeline and tests each ingredient under a strict spatial split. We ask five research questions:

RQ1 Does night-light steering improve municipal GDP estimates from satellite embeddings?

RQ2 Does aligning the embeddings with generated text (captions plus data-derived fact sentences) help on top of steering?

RQ3 Do points of interest and road lengths add information on top of text and steering?

RQ4 Does a hybrid with the night-light benchmark beat the best single model?

RQ5 Where and why does the best model fail?

Contributions. (i) A reproducible 1 km dataset for Minas Gerais joining GSED, VIIRS, ESA WorldCover, OpenStreetMap, Microsoft building footprints, 10,000 captioned Esri tiles and IBGE statistics. (ii) An ablation of twelve model variants on identical spatial folds, with block-bootstrap intervals for each research question. (iii) A caption-cleaning step that checks generated sentences against map data, and a blind audit of its effect. (iv) An error analysis that identifies where satellite data cannot see the economy.

2 Study area and data

Why Minas Gerais. Brazil's statistics office IBGE publishes GDP for every municipality, and the 2022 Census gives exact population counts. Minas Gerais has 853 municipalities, far above the 150-unit minimum of the proposal, and one of the widest income ranges in the country: GDP per capita runs from R$ 8,049 in São João das Missões (Jequitinhonha–north) to R$ 390,991 in Catas Altas (an iron-mining town), a ratio of 49. The target is log(GDP/population), with mean 10.14 and standard deviation 0.60; on this scale an error of 0.1 is about 10%. Figure 1 shows its distribution and map.

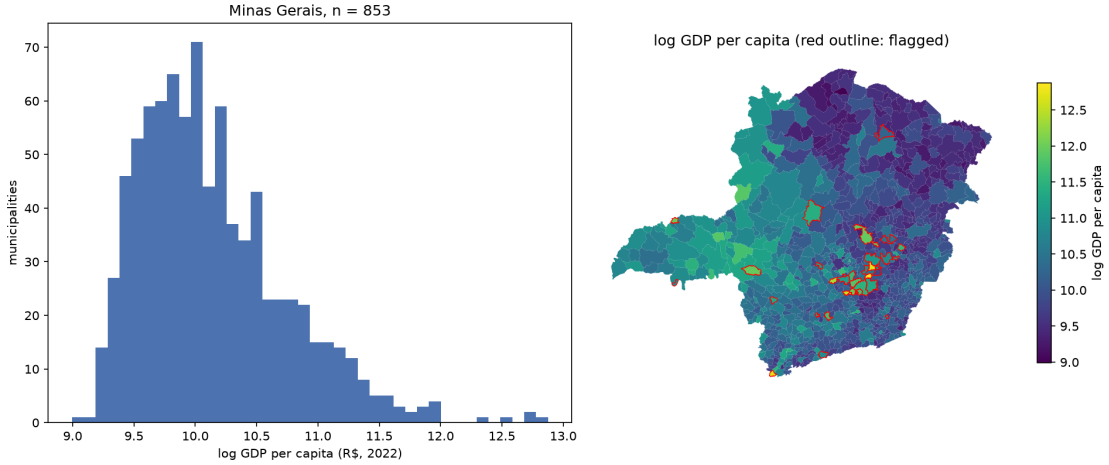


Figure 1: Official 2022 log GDP per capita of the 853 municipalities (IBGE). The south-west (Triângulo) and the Belo Horizonte metro are rich; the north and north-east (Jequitinhonha, Norte de Minas) are poor.

Grid and cells. The state is covered by a 1186×963 grid of 1 km cells in SIRGAS 2000 / UTM 23S; 586,873 cells have their centre inside a municipality. GSED, VIIRS and WorldCover are read directly at 1 km from Earth Engine (GSED at 10 m in the grid CRS, then averaged). We drop 1,441 cells that are $\geq$90% water and 10,266 cells that are $\geq$95% tree or bare ground *and* have no built-up area, light, POI or road. The remaining 575,166 cells carry the 64 GSED numbers, log night light $\log(1 + \text{NTL})$, land-cover shares, POI counts and road lengths (Table 1). Night lights are never used as a GDP proxy target: the target is always official IBGE GDP.

3 Method

3.1 Text: captions and fact sentences

We sample 10,000 kept cells, 2,000 per fold, stratified by built-up tercile and night-light decile so that towns are oversampled relative to pasture. For each we download an Esri World Imagery crop of the cell and caption it with Qwen2.5-VL-7B-Instruct on Kaggle (2×T4, fp16, temperature 0.2, at most 200 new tokens). A 200-tile pilot showed accurate but long captions (median 142 words, 12% cut off at the token limit), so the final prompt asks for *at most 5 short sentences, under 90*

Table 1: Data layers. All are free; all raster layers are averaged onto the same 1 km grid (EPSG:31983).

Layer	Source (year)	Role
Satellite embeddings	Google GSED (Earth Engine, 2022), 64 bands, 10 m	Main image input
Night lights	VIIRS DNB annual V2.2, `average_masked` (2022)	Stage B proxy label; NTL benchmark
Land cover	ESA WorldCover v200 (2021)	Masking, fact sentences, caption checks, baseline V2
POIs, roads, waterways	OpenStreetMap, Geofabrik extract of 1 Jan 2023	POI counts (10 classes), road km (4 classes)
Buildings	Microsoft Global ML Building Footprints (10.0 M) + OSM (0.36 M)	Caption checks
Caption imagery	Esri World Imagery, zoom 17 (≈1.1 m/px), 768×768 px	VLM input only
GDP, value added	IBGE PIB dos Municípios (GDP 2022; sector shares 2021)	Stage C target; error analysis
Population	IBGE Census 2022	GDP per capita
Boundaries, regions	IBGE Malha 2022; Regiões Geográficas Imediatas (2017)	Cell assignment; spatial blocks

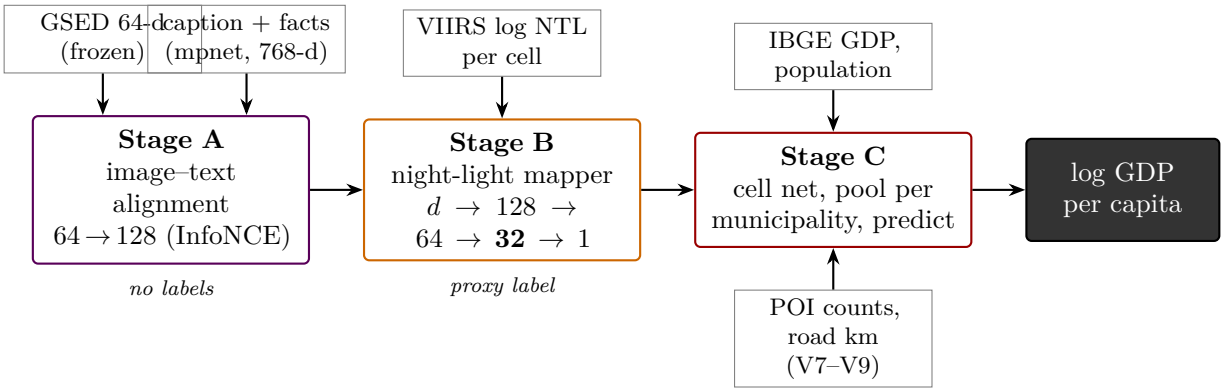


Figure 2: Pipeline. Stage A needs text only in training; at prediction time each cell enters through its embedding alone. Variants skip stages (Table 2); all stages are retrained in every cross-validation round.

words, covering built-up area and buildings, roads, vegetation and farmland, water, and industry or mining; only what is clearly visible; no place names; no summary sentence.

Each cell also gets 2–4 *fact sentences* written from its data, e.g. "Mostly built-up: 74% built-up, 24% trees. 3 schools and 1 public building are mapped. 2.6 km of medium roads and 14.2 km of minor roads." Night lights are deliberately left out of the facts so that the effects of text (RQ2) and steering (RQ1) stay separable.

Cleaning. Captions are split into 60,023 sentences. Pass 1 removes filler ("the overall scene suggests…"). Pass 2 checks claims against the cell's data with a keyword lexicon: water words need water cover or a mapped waterway; building words need built-up cover, buildings or POIs; industry or mining words need an industry POI or enough built-up area; farm, forest, road and urban words have similar checks. Sentences with a negation ("no river is visible") skip the check. Pass 3 scores each tile against its cleaned caption with RemoteCLIP ViT-B/32 and drops captions below the 5th percentile of a seeded 200-caption subset (246 captions). In total 84% of sentences are kept; the removals are filler (4,183), industry (1,130), building (1,108), water (1,046), road (1,039), farm (732), urban (66) and forest (35). Three text versions feed the ablations: `full` (cleaned caption + facts), `raw` (uncleaned caption) and `facts` (no VLM at all).

3.2 Stage A: image–text alignment

GSED vectors are standardised with training-cell statistics. A two-layer projector maps the 64 GSED numbers to a 128-d unit vector ($64 \to 256 \to 128$, GELU, dropout 0.1); a second projector maps the frozen 768-d `all-mpnet-base-v2` sentence embedding of the text to the same space. A symmetric InfoNCE loss with learnable temperature (initial 0.07) pulls each cell towards its own text. We train on the ≈6,000 captioned cells of the training municipalities (AdamW, lr 10^{-3}, batch 256, early stopping on the validation fold's captions) and then embed *every* kept cell with the image projector. GSED is frozen, so Stage A can only reorganise the information GSED already holds; it cannot add new information.

3.3 Stage B: night-light steering

A network $d \to 128 \to 64 \to 32 \to 1$ (ReLU) predicts each cell's standardised $\log(1 + \mathrm{NTL})$ from its embedding (raw GSED, $d = 64$, or a Stage A embedding, $d = 128$), trained on all training-fold cells and early-stopped on the validation fold. The 32-number last hidden layer $\mathbf{z}$ is the steered embedding. The embedding paper does not publish its layer widths; ours follow its three-hidden-layer shape.

3.4 Stage C: from cells to municipalities

Each cell's input (z, a Stage A embedding or raw GSED, plus optionally $\log(1 + x)$ POI counts and road km) passes a per-cell network (hidden 32 or 64, dropout 0.1, softplus output of 16 numbers). Outputs are pooled per municipality (log-sum or mean), a municipal network $16 \to 16 \to 1$ predicts log total GDP, and log population is subtracted. Training is full-batch on log GDP per capita (Adam, lr 10^{-3} halved after 50 epochs without improvement, early stopping on a 100-epoch rolling mean of validation MSE, at most 3,000 epochs). The output bias starts at the training mean, without which Adam could not reach the target scale in time. A 12-setting grid (hidden $\{32, 64\} \times$ weight decay $\{10^{-4}, 10^{-3}, 10^{-2}\} \times$ pooling {log-sum, mean}) is tuned on the validation fold, and the chosen setting is retrained with 5 seeds whose mean is the prediction. Log-sum pooling won in 33 of 35 variant–rounds. Finally a linear recalibration (official on predicted) is fitted on the validation fold and applied to test.

3.5 Baselines, hybrid and variants

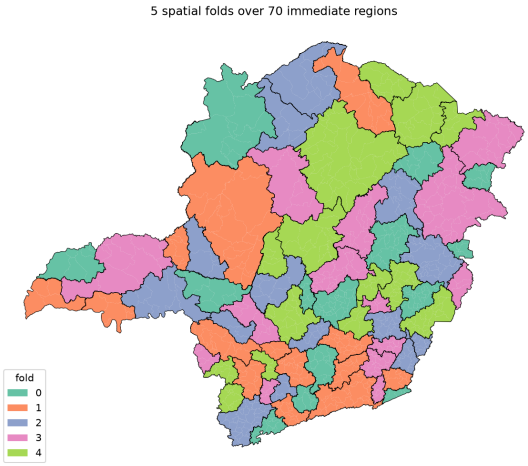
V1 is the night-light benchmark of Hu and Yao (2022) and the embedding paper: OLS of log GDP per capita on $\log(1 + \sum \mathrm{NTL}) - \log(\mathrm{pop})$, linear or quadratic chosen on validation. V2 is a ridge on mean land-cover shares, log cell count and log population. V11 is a ridge on mean embeddings (each embedding type separately) and V12 LightGBM on the same municipal features. V10 blends the best of V3–V9 (by validation RMSE) with V1: $\alpha\, \hat{y}_{\mathrm{model}} + (1 - \alpha)\, \hat{y}_{\mathrm{V1}}$, $\alpha \in \{0, 0.25, \ldots, 1\}$ chosen on validation.

3.6 Evaluation

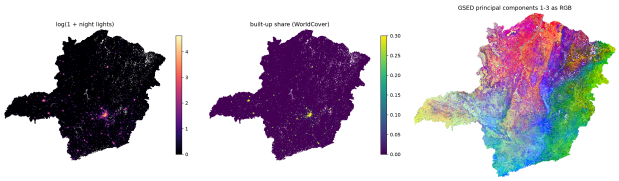
The 70 IBGE immediate regions are the spatial blocks. Of 10,000 random assignments of regions to 5 folds we keep the one most balanced in municipality count, mean and spread of log GDP per capita and urban share (Figure 3a; 160–184 municipalities and 14 regions per fold). In round r, fold r is the test fold, fold $r{+}1$ validates, and the other three train. Test municipalities never influence any stage, scaler, threshold, hyperparameter, hybrid weight or caption used in training; captions of test cells are never loaded. We pool the 853 test predictions and report R^2, RMSE and MAE. For each research question we compute the RMSE difference with a 95% block bootstrap that resamples the 70 regions 2,000 times; an interval containing 0 is reported as "no clear effect".

Table 2: Model variants. All use the same five rounds; A = Stage A, B = Stage B, C = Stage C.

ID	Pipeline	Question
V1	night lights only (OLS)	benchmark
V2	land-cover shares → ridge	is a simple model enough?
V3	raw GSED → C	base value of embeddings
V4	raw GSED → B → C	RQ1: steering (reproduces the embedding paper)
V5	A(full) → C	text without steering
V6	A(full) → B → C	RQ2: text + steering
V7	A(full) → B → C + POI/roads	RQ3: full model
V8	as V7, uncleaned captions	value of cleaning
V9	as V7, fact sentences only	is the VLM needed?
V10	best of V3–V9 blended with V1	RQ4: hybrid
V11	ridge on mean embeddings	simple check of each embedding
V12	LightGBM on municipal features	strong off-the-shelf baseline



(a) Five spatial folds over the 70 immediate regions.



(b) Quality check: log night light, built-up share, GSED principal components as RGB.

Figure 3: Spatial cross-validation design and the cell table.

4 Results

4.1 Main results

Table 3 gives the main results. Every Stage C model (V3–V10) beats the night-light benchmark by a wide margin (R^2 0.44–0.52 vs. 0.27), and also beats ridge and LightGBM on the same embeddings (0.35–0.40). The deep pooling therefore earns its place: learning which cells matter, rather than averaging them, is worth about 0.1 in R^2. The best model is the hybrid V10 (R^2 0.523, RMSE 0.412); without the 34 flagged municipalities its RMSE drops to 0.381. Figure 4 shows its predictions against official values next to V1's: V1 squeezes almost all municipalities into a narrow band, while V10 spreads along the diagonal but still pulls the extremes towards the middle. Other ensemble variants are reported in the repository (`reports/tables/main_results.csv`).

4.2 Research questions 1–4

RQ1, steering. Steering raw GSED with night lights does not help: V4 is slightly worse than V3 and the interval is centred on zero. This does *not* reproduce the embedding paper's gain. Stage B itself works (validation R^2 0.74–0.85 for log night light; Table 5), so the steered space does encode light, but the 32-number bottleneck discards information that Stage C could otherwise use, and light is already partly recoverable from GSED.

Table 3: Out-of-sample results, all 853 municipalities pooled over 5 rounds (log GDP per capita). "Unflagged" excludes the 34 municipalities flagged for industry share above 60% or top-1% GDP per capita. Calibration slope: regression of official on predicted (1 = no bias).

ID	Model	R^2	RMSE	MAE	RMSE sd (rounds)	RMSE unflagged	calib. slope
V1	night lights only	0.271	0.510	0.406	0.015	0.477	0.97
V2	land cover → ridge	0.124	0.559	0.373	0.155	0.508	0.60
V3	GSED → C	0.459	0.439	0.323	0.020	0.408	0.91
V4	GSED → B → C	0.441	0.446	0.333	0.054	0.406	0.84
V5	A(full) → C	0.483	0.429	0.313	0.021	0.391	0.90
V6	A(full) → B → C	0.501	0.422	0.308	0.019	0.378	0.94
V7	V6 + POI/roads	0.502	0.421	0.306	0.029	0.373	0.96
V8	V7, uncleaned captions	0.501	0.422	0.309	0.048	0.380	0.93
V9	V7, facts only	0.483	0.429	0.318	0.049	0.385	0.92
V10	**hybrid (best + V1)**	**0.523**	**0.412**	**0.302**	0.035	0.381	0.95
V11	ridge, mean A(full) emb.	0.397	0.463	0.330	0.038	0.395	0.90
V11	ridge, mean raw GSED	0.345	0.483	0.346	0.064	0.416	0.90
V12	LightGBM	0.386	0.468	0.335	0.047	0.401	0.96

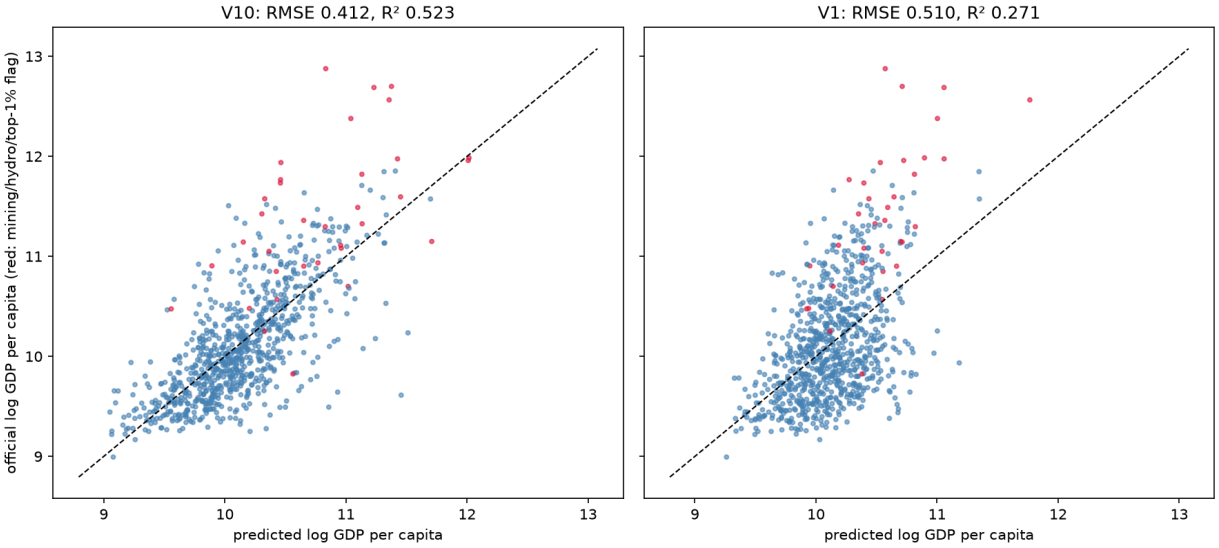


Figure 4: Predicted vs. official log GDP per capita, test predictions of all five rounds. Left: hybrid V10; right: night lights only (V1). Red: flagged mining, hydro or top-1% municipalities.

Table 4: Research-question tests: RMSE difference (first minus second model) with 95% block-bootstrap interval over the 70 regions. Negative = first model better.

Question	Comparison	ΔRMSE [95% CI]	Verdict
RQ1 night-light steering	V4 vs V3	+0.007 [−0.028, +0.041]	no clear effect
RQ2 text + steering	V6 vs V4	−0.025 [−0.050, +0.004]	no clear effect (borderline)
RQ3 POIs and roads on top	V7 vs V6	−0.001 [−0.017, +0.016]	no clear effect
RQ4 hybrid vs best single	V10 vs V8	−0.010 [−0.035, +0.015]	no clear effect

RQ2, text. Aligning with text is the largest single gain: V6 improves on V4 by 0.025 RMSE (about 2.5% in GDP per capita), and V5 improves on V3 by 0.010 without steering. The interval only just includes zero. The ablations point the same way: uncleaned captions (V8) and fact sentences alone (V9) are no better than the cleaned captions plus facts (V7), so the gain does not hinge on the vision–language model or on cleaning.

RQ3, POIs and roads. Adding POI counts and road lengths as numbers changes nothing (V7 vs V6). The fact sentences already mention POIs and roads, so their information reaches the model through text; OSM is also sparse in rural Minas, and the OSM-completeness term has no effect on errors (Section 5).

RQ4, hybrid. The hybrid is the best model overall, but its edge over the best single model chosen on validation (V8) is small and not significant. In three rounds the validation-optimal weight was $\alpha = 1$ (no night lights), in two $\alpha = 0.75$; the base model also differs by round (V6, V5, V8, V3, V3). Much of V10's advantage is thus picking the best Stage C model per round on validation rather than blending.

A lesson on honest re-testing. The first complete run (before Stage C was revised) gave a different picture: V3 RMSE 0.555, V4 0.491, and RQ1 "V4 better" with an interval excluding zero (-0.064 [-0.115, -0.019]). Inspecting *validation* curves showed that Stage C predictions were over-dispersed and that plain sum pooling never won. Adding the validation recalibration and mean pooling, and replacing sum pooling in the grid, cut V3's RMSE by 21% and V4's by 9%, and the apparent steering effect vanished: steering had been compensating for a poorly calibrated aggregator. All changes were chosen on validation data only, but we made them after seeing the first test results, which we report here as a researcher degree of freedom.

4.3 Did each stage do its job?

Table 5: Stage diagnostics, mean over 5 rounds (validation fold). Stage A: image→text recall@k within groups of 256 validation pairs (chance recall@10 ≈ 4%) and a linear probe from the embedding to built-up share. Stage B: validation R^2 and RMSE of standardised log night light; 81% of training cells are unlit.

Stage A					Stage B		
Text	R@1	R@5	R@10	probe R^2	Input	R^2	RMSE
full (cleaned + facts)	0.123	0.361	0.522	0.67	raw GSED	0.775	0.176
raw caption	0.065	0.229	0.349	0.68	A(full)	0.763	0.181
facts only	0.122	0.376	0.551	0.64	A(raw)	0.761	0.182
					A(facts)	0.752	0.185

Stage A aligns image and text far above chance: the right text ranks among the top 10 of 256 candidates for 52% of validation cells with the full text, against about 4% by chance (Table 5). Raw captions align worst (35%) and fact sentences best (55%), which is expected: the facts are computed from land cover, roads and POIs, which GSED can see, while the free-form captions also contain sentences GSED cannot match. The embedding keeps basic land information (probe R^2 0.64–0.68 for built-up share). Stage B predicts night light well from any embedding, best from raw GSED, consistent with Stage A only reshaping GSED.

Figure 5 repeats the embedding paper's neighbour check. In the steered space z, a cell's nearest neighbours are 2–4 times farther away (e.g. 82 vs 292 km for a small town) but more alike in night light for the small town (log-NTL gap 0.66 → 0.46) and the mining cell (0.90 → 0.50), and equal for the industrial and Belo Horizonte cells. Steering does what the paper describes, turning "looks like its neighbours" into "is as active as", yet this did not translate into better municipal GDP here (RQ1).

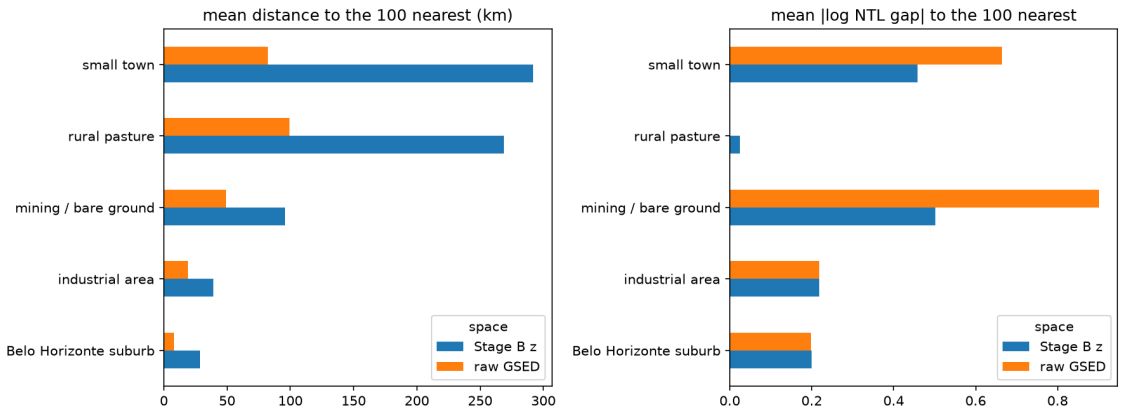
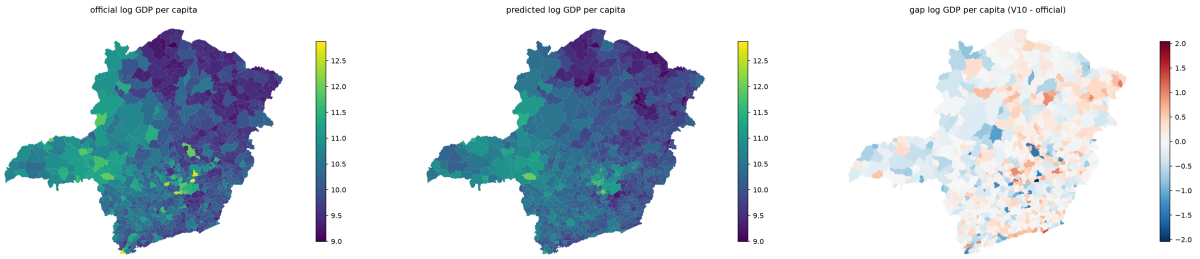


Figure 5: The embedding paper's neighbour check, round 0. For five reference cells, the 100 nearest cells by cosine similarity in raw GSED and in the steered space z: mean geographic distance (left) and mean absolute gap in log night light (right).

5 Where and why the model fails (RQ5)



(a) Official **(b)** Predicted (V10) **(c)** Gap (V10 − official)

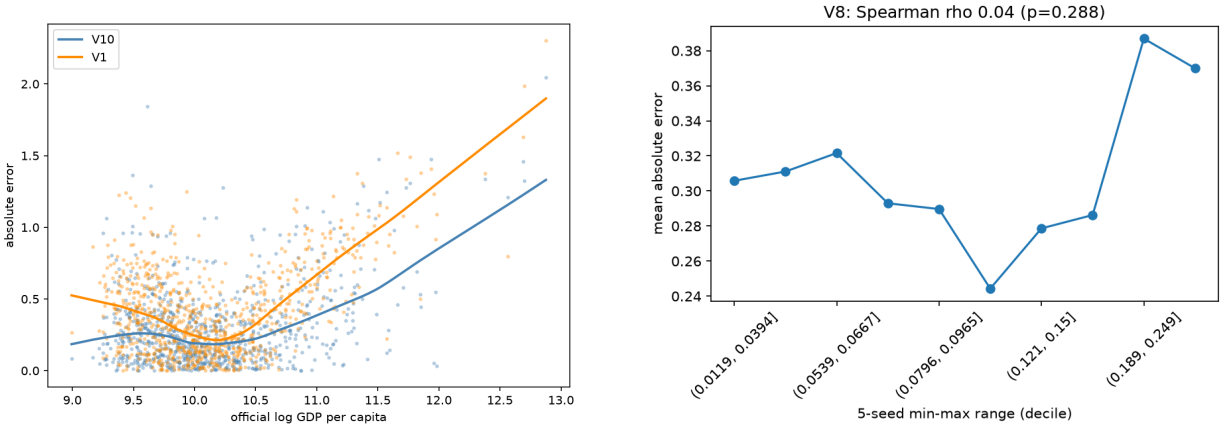
Figure 6: Municipal maps of log GDP per capita, same colour scale for official and predicted. Blue: under-predicted; red: over-predicted.

The predicted map reproduces the main geography (Figure 6): the rich Triângulo and south, the Belo Horizonte metro, and the poor north-east. The gaps are spatially clustered: Moran's I of the test residuals over neighbouring municipalities is 0.16 for V10 ($p = 0.001$), half the 0.33 of night lights alone. The embedding model removes much of the regional bias of night lights, but a regional factor remains.

Table 6: Absolute error of V10 and V1 regressed on municipality traits (OLS, standard errors clustered by immediate region, $n = 853$). Sector shares are from 2021.

Term	V10 hybrid coef. (SE)	p	V1 night lights coef. (SE)	p
official log GDP per capita	0.081 (0.035)	0.02	0.190 (0.037)	<0.001
built-up share	0.836 (0.473)	0.08	0.791 (0.250)	0.002
services share of value added	−0.285 (0.132)	0.03	−0.332 (0.123)	0.007
industry share of value added	0.185 (0.121)	0.13	−0.107 (0.102)	0.29
flagged (mining/hydro/top 1%)	0.165 (0.090)	0.07	0.272 (0.079)	<0.001
OSM completeness	−0.018 (0.082)	0.82	−0.099 (0.054)	0.07

Errors grow with income, less than for night lights. Table 6 mirrors the embedding paper's error regression. IBGE publishes no per-municipality data-quality score, so we analyse income and economic structure instead. For both models the absolute error rises with official income, but the slope for V10 is less than half that of V1 (0.08 vs 0.19 per log point), and the LOWESS curves in Figure 7a show the same. Errors are smaller where services dominate the economy and larger in flagged municipalities, though for V10 the flag is only weakly significant. OSM completeness does not matter for either model.



(a) Absolute error vs. official income, LOWESS.

(b) Ensemble spread vs. error (V8, 5 seeds).

Figure 7: Error structure and uncertainty.

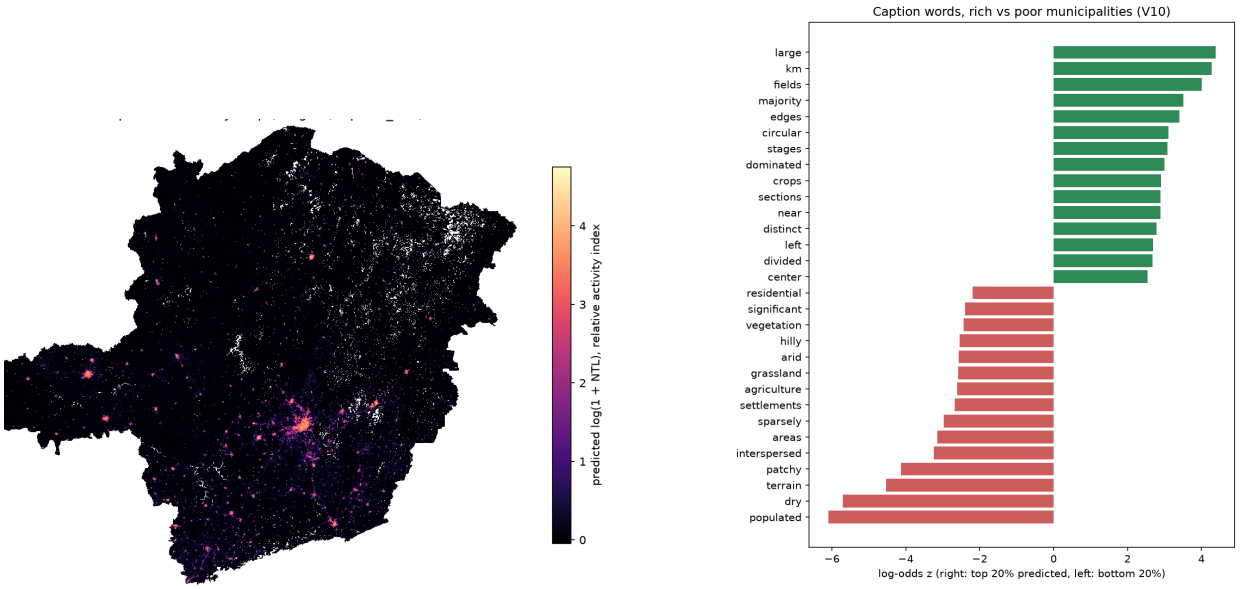
The ten largest gaps. Seven of the ten largest errors are under-predictions of mining or heavy-industry municipalities (Table 7). Catas Altas, Conceição do Mato Dentro, São Gonçalo do Rio Abaixo and Itatiaiuçu host large iron-ore mines, Ouro Branco a large steel mill, and Extrema a logistics and industrial hub on the São Paulo highway. In each, a handful of cells produce most of the output, while the sampled tiles show forest, pasture and rock, and their captions say "no industrial activity is visible". Satellites cannot see a mine's revenue, as the proposal anticipated. The three over-predictions are small towns with large public-sector shares (39–48%): Santa Cruz de Minas, one of Brazil's smallest municipalities and almost entirely urban next to São João del-Rei, and Mário Campos, a Belo Horizonte commuter town, look as built-up as richer places but residents earn elsewhere. These reasons are candidates from sector shares and public knowledge, not proof.

Table 7: The ten largest gaps of V10 (predicted minus official log GDP per capita) and candidate reasons.

Municipality	official	gap	industry	candidate reason
Catas Altas	12.88	−2.04	81%	iron-ore mining
Santa Cruz de Minas	9.61	+1.84	8%	tiny urban commuter town, public sector 42%
Ressaquinha	11.51	−1.48	50%	single large plant (not identified)
Conceição do Mato Dentro	11.94	−1.48	80%	iron-ore mine and pipeline
São Gonçalo do Rio Abaixo	12.69	−1.46	78%	iron-ore mine
Mário Campos	9.49	+1.37	8%	Belo Horizonte commuter town
Itatiaiuçu	12.38	−1.34	80%	iron-ore mining
Extrema	12.70	−1.32	26%	logistics and industry hub
Ouro Branco	11.77	−1.31	72%	integrated steel mill
Chiador	9.64	+1.29	3%	small public-sector economy (48%)

Uncertainty. The spread of the 5-seed ensemble does not track the error (Spearman $\rho = 0.04$, $p = 0.29$; Figure 7b). Seed variation measures optimisation noise, not the systematic error of a mining town, so it should not be used as a confidence interval.

A 1 km activity map. Because every municipality is tested exactly once, we can assemble an out-of-sample map of Stage B's predicted night light for every cell, each taken from the round in which its municipality was in the test fold (Figure 8a). It is a relative activity index for display, not a GDP map.



(a) Out-of-sample 1 km activity index (Stage B). **(b)** Caption words, top vs bottom 20% predicted.

Figure 8: Cell-level output and what the captions say.

6 What the text says

Nearest captions. For a few municipalities we list the training captions nearest to the municipality's mean cell embedding in the Stage A space of the round where it is tested (`reports/tables/explanations.md`). For Belo Horizonte (official 10.82, predicted 10.91) all five are dense residential grids with red-tiled roofs and 60–74% built-up. For the poorest municipality, São João das Missões (8.99 vs 9.08), they describe sparse settlements in dry scrub. For Conceição do Mato Dentro, a mining town (11.94 vs 10.46), they describe forest and pasture, which is exactly why it is under-predicted: the text describes what is visible, and the mine's value is not visible in most of the municipality's cells.

Rich vs poor words. Figure 8b contrasts caption words of the top and bottom 20% of predicted municipalities (log-odds with an informative Dirichlet prior, Monroe et al., 2008). Rich places are described with *large, fields, circular, crops, divided, sections*: mechanised agriculture with centre-pivot irrigation, the Triângulo's agribusiness. Poor places get *sparsely populated, dry, terrain, patchy, arid, hilly, settlements*. The words are interpretable and match the economic geography of the state.

6.1 How reliable are the captions? A blind audit

We drew 300 captions at random and labelled each of their 1,801 sentences as *correct*, *false* (claims something the tile contradicts) or *vague* (true but says nothing specific), looking at the tile and without seeing which sentences the cleaning had dropped. Raw and cleaned results are derived from the same labels. **The labelling was done by an LLM (Claude, Anthropic), looking at each tile beside its sentences under fixed written rules, not by a human annotator.** A second, human labeller for a subset (to compute Cohen's kappa) was not available in time; the label sheet is in the repository for that purpose.

Table 8: Caption audit, 300 random captions (1,801 sentences), Wilson 95% intervals. Labels: 1,440 correct, 235 vague, 126 false.

Measure	n	value	95% CI
captions with ≥1 false claim, raw	300	34.7%	[29.5, 40.2]
captions with ≥1 false claim, after cleaning (captions still used)	284	23.2%	[18.7, 28.5]
rule-dropped sentences that were false or vague (precision)	288	58.0%	[52.2, 63.5]
false sentences removed by cleaning (recall)	126	41.3%	[33.1, 50.0]

Only 7% of sentences are false, but because captions have six sentences, a third of raw captions contain at least one (Table 8). Cleaning cuts this to 23%. The typical false claims are: "no buildings" on tiles with a few farmhouses, "farmland" on natural scrub or forest, "industrial or mining activity" on centre-pivot fields or quarries, "no roads" on tiles with dirt tracks, and ponds that are not there. The cleaning removes 41% of the false sentences, and 58% of what it removes is false or vague; the rest are true sentences the keyword rules misjudged (for example "field" in a town with a sports field). Consistent with V8 ≈ V7, this level of noise does not hurt the GDP estimate, probably because the alignment only uses text in training and the fact sentences carry most of the reliable content.

7 Limitations

- **Frozen GSED.** Stage A and B can only reorganise the 64 GSED numbers; any gain from text comes from a better-organised space, not new information. Fine-tuning an image encoder on the tiles was out of scope.
- **Contrastive loss only.** UrbanCLIP's ablation found the language-modelling loss alone beat contrastive alone; we kept only the contrastive part, as in the proposal.
- **Audit labels.** The 300-caption audit was labelled by an LLM, not by people, and by a single labeller, so no agreement statistic is available. LLM judgements of small features (a pond, a farmhouse) can be wrong.
- **Revisions after the first test results.** Stage C's recalibration and grid were chosen on validation data, but after the first test results were seen (Section 4.2).
- **Data years and sources.** WorldCover is 2021 and sector shares are 2021; Esri imagery mixes capture dates around 2020–2023; OSM is sparse in rural areas. IBGE publishes no data-quality indicator per municipality, so that part of the embedding paper's error analysis could not be repeated.
- **Hybrid.** V10 can pick a different base model in each round, so it is a procedure rather than one model.
- **Growth extension not done.** The optional growth test (Phase 10) needs GSED and VIIRS for 2018–2021; the Earth Engine project's monthly quota was exhausted, and the gate rule of the plan dropped it first.
- **One state.** All results are for Minas Gerais; transfer to other states or countries is untested.

8 Conclusion

Free satellite data explain about half of the variation in municipal GDP per capita across Minas Gerais when tested on regions the model never saw (R^2 0.52 vs 0.27 for night lights alone). The gain comes mainly from learning which 1 km cells matter (Stage C pooling) and, to a smaller degree, from aligning embeddings with text. Night-light steering, POIs and roads, and the hybrid add

nothing significant once Stage C is well calibrated, and an early "steering helps" result disappeared when the aggregator was fixed, which shows how much honest spatial testing matters. The model fails where output is concentrated and invisible: mines, steel mills and logistics hubs, together with small commuter and public-sector towns. Generated captions are useful but noisy (a third contain a false claim), and simple data checks remove a good part of that noise. A natural next step is to give the model information that satellites cannot see, such as registered firms or mining licences, for the municipalities where it fails.

Reproducibility. Code, configuration and all tables and figures are in the repository (tag `v1.0-results`). `scripts/run_all.sh` reruns every step from the raw downloads; data live outside the repository. The release folder contains the cell feature table, the cleaned captions, the audit labels, all predictions and the 1 km activity GeoTIFF; Esri tiles are not redistributed.

References

Brown, C. F. et al. (2025). AlphaEarth Foundations / Google Satellite Embedding dataset (GSED), Earth Engine.

Hu, Y. and Yao, J. (2022). Illuminating economic growth. *Journal of Econometrics* 228(2), 359–378.

IBGE (2024). Produto Interno Bruto dos Municípios; Censo Demográfico 2022; Divisão Regional do Brasil 2017.

Monroe, B. L., Colaresi, M. P. and Quinn, K. M. (2008). Fightin' words. *Political Analysis* 16(4), 372–403.

Liu, F. et al. (2024). RemoteCLIP: a vision language foundation model for remote sensing. *IEEE TGRS* 62.

Bai, S. et al. (2025). Qwen2.5-VL technical report. arXiv:2502.13923.

Yan, Y. et al. (2024). UrbanCLIP: learning text-enhanced urban region profiling with contrastive language–image pretraining from the web. *Proceedings of the ACM Web Conference 2024*.

Yue, Zhao and Hu (2026). Estimating economic activity from satellite embeddings. *Applied Sciences* 16, 582.

Zanaga, D. et al. (2022). ESA WorldCover 10 m 2021 v200.